Pre-registration: Studies 6, 7, 8 + judge audit (graphical-reasoning, AAAI 2027)

Silvere Gangloff

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Authored: 2026-05-21. **Repo state at authoring:** commit `9e5fd98` on branch `main`. **Submitter:** Silvere Gangloff. **Target venue:** AAAI 2027 main track. **Authoritative working document:** `paper/publication_plan.md`.

Evidence trail: this file's git history. The commit that first introduced each hypothesis (`paper/pre_registration.md` was created at commit `5eb78a4`, 2026-05-21, and last substantively revised at commit `1cfc1f2`, 2026-05-23 — both before Study 6 data was collected) is the timestamp that defeats post-hoc-framing concerns. This is a weaker guarantee than a third-party timestamping service (OSF, arXiv): the author controls the git history. The author commits to *not rewriting* commits that touch this file once data collection for the corresponding study has begun.

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1. Purpose of this document

This document fixes the hypotheses, primary outcomes, statistical tests, decision rules, and stopping rules for **three new studies (Studies 6, 7, 8) and the human judge audit (HJ)** *before any data for those studies is collected*. It exists to defeat post-hoc-framing concerns at peer review.

It does **not** retroactively pre-register Studies 1–5. Those are exploratory by definition and the paper will say so. The replication of Study 5 at higher n (re-named *Study 6* below) is pre-registered here.

2. Transparency: what is and isn't pre-registered

Study	Pre-registered?	Status as of 2026-05-21
1 — A/B sweep (120 attempts)	No (exploratory)	Complete, committed in main
2 — Input-framing probe (27)	No (exploratory)	Complete, committed
3 — Forced-loop ablation (54)	No (exploratory)	Complete, committed
4 — Recall probes 4a, 4a' (36)	No (exploratory)	Complete, committed
5 — Adversarial pilot (48)	No (exploratory)	Complete, committed
6 — Scaled adversarial (96 new)	**Yes (this doc)**	Not yet run
7 — Path-only regression on inline `math-viz` (144)	**Yes (this doc)**	Not yet run

Study	Pre-registered?	Status as of 2026-05-21
8 — Geometric adversarial corpus (240)	**Yes (this doc)**	Not yet run
HJ — Human judge audit (120 cells)	**Yes (this doc)**	Not yet run

3. Hypotheses, tests, and decision rules

H6 — Modality equivalence on the original adversarial corpus

Hypothesis (declarative): On the 4-problem false-claim corpus, at combined sample size n = 8 (existing Study 5) + 16 (new Study 6) = 24 runs per condition, the three forced-verification conditions (`forced_visual`, `forced_textual`, `forced_reflection`) have indistinguishable false-claim catch rates.

Primary outcome: count of `false`-variant runs graded `CORRECT` by the LLM judge (verdict = `in-correct` *and* canonical-answer regex match), per (condition $\times$ problem), summed across problems.

Test: Fisher's exact on the 3 $\times$ 2 contingency (condition $\times$ {caught, not caught}) over all `false`-variant cells. Pairwise Fisher's exact (3 pairs) with Bonferroni correction (α effective = 0.05/3 = 0.0167).

Decision rule:

- *Modality-doesn't-matter supported* iff no pairwise comparison reaches p < 0.0167 *and* the overall 3 $\times$ 2 Fisher's exact yields p > 0.05.
- *Modality matters* iff any pairwise comparison reaches p < 0.0167.

No-peeking commitment: Study 6's 96 new cells are run before any hypothesis test is computed. The cell-by-cell grades are computed only after all 96 runs are recorded.

Study 7 design note (revised 2026-05-23)

The original Study 7 design proposed *adding* an inline-content mode to a path-returning `math-viz`. That design is moot: upstream `math-viz` already returns inline image content as of commit `1622bac` on 2026-05-07, and Studies 2–5 in the long-form paper all ran post-switch. Study 7 is therefore reframed as a **path-only regression.** We build a thin wrapper around the current `math-viz` that strips the inline image content block from each tool result, leaving only the filesystem path as text. The baseline arm is the unmodified upstream tool (inline image returned); the manipulated arm is the wrapped tool (path-only). Hypothesis direction and sample size are unchanged from the pre-revision draft.

H7a — Inline-image return lifts voluntary adoption

Hypothesis: In the `optional` $\times$ `inline` arm of Study 7, voluntary `math-viz` adoption (fraction of cells with $\geq$1 successful `math-viz` invocation) is $\geq$ 15 absolute percentage points higher than in the `optional` $\times$ `path-only` arm.

Primary outcome: binary adoption per cell.

Test: Paired binomial across the 6 problems (`optional` $\times$ `inline` vs `optional` $\times$ `path-only` adoption rates per problem). Test statistic: one-sided sign test on the per-problem difference; α = 0.05.

Decision rule:

- *H7a supported* iff the one-sided sign test reaches $p < 0.05$ *and* the mean per-problem lift is $\geq$ 15 absolute points.

H7b — Inline-image return shifts grounding from metadata to image

Hypothesis: Conditional on adoption, agents in the inline arm reference image features (e.g., shape, curve crossing, slope visible, region position) in $\geq$ 50% of emitted verdicts; agents in the path-only arm reference image features in $\leq$ 20% of emitted verdicts.

Caveat from prior data. §9.2 of the long-form paper (Studies 2–5, post-switch, `forced_visual` arm) finds that even with inline images delivered automatically, the agent grounds verdicts in the plotting tool's *numerical metadata* rather than in image features. Those cells were *forced* uses; Study 7 measures *voluntary* uses, where the prior probability of image grounding may differ. We pre-register the directional hypothesis but report both possibilities — *inline lifts grounding* (H7b supported) and *neither arm grounds in images* (H7b refuted in an informative direction) — as primary findings of the study.

Primary outcome: binary "image-feature reference" per verdict, hand-coded by the author on 20 random adopted cells per arm (40 total) using the fixed coding rubric below, committed before any hand-coding begins.

Test: Fisher's exact on the 2 $\times$ 2 (arm $\times$ {image-feature, no}) table; α = 0.05.

Decision rule:

- *H7b supported* iff Fisher's exact $p < 0.05$ *and* the empirical proportions match the directional hypothesis.
- *H7b refuted (informative null)* iff both arms have $\leq$ 25% image-feature reference rate. This is the §9.2-consistent outcome and is reported as such.

Coding rubric (committed before coding):

- An "image-feature reference" requires the verdict text to invoke a *visual* property (shape, curvature visible, region containment read off the plot, crossing point identified by eye) that could not be obtained from numerical sample values alone.
- A "metadata-only reference" cites numbers (e.g., "f(1) = 0.24"), symbolic results, or tool textual return without invoking a visual property.
- Ambiguous: flagged separately; counted as "no image-feature reference" for the primary test, audited in the appendix.

H8 — Picture beats numerics on the geometric adversarial corpus

Hypothesis: On the 8-problem geometric adversarial corpus, `forced_visual` catches false claims at a strictly higher rate than `forced_textual` (one-sided).

Primary outcome: per-problem false-claim catch rate.

Test: Paired one-sided binomial across the 8 problems (per-problem catch rate, `forced_visual` — `forced_textual`). Test statistic: sign test on per-problem differences; $\alpha = 0.05$. Secondary: paired comparison `forced_visual` vs `forced_reflection` (two-sided, $\alpha = 0.05$).

Decision rule:

- *H8 supported* iff one-sided sign test $p < 0.05$ in the `forced_visual` > `forced_textual` direction *and* the mean per-problem advantage is ≥ 10 absolute points.
- *Null result* otherwise. A null result is itself a publishable finding — see decision-gate 3 in `publication_plan.md`.

HJ — Human judge audit calibration

Hypothesis: Cohen's κ between human grades and LLM judge grades on the 120-cell stratified random sample is ≥ 0.6 overall and ≥ 0.5 per study stratum.

Primary outcome: κ statistic, computed over the binary {CORRECT, not-CORRECT} judgment (treating UNCLEAR as not-CORRECT for the primary test; sensitivity analysis treating UNCLEAR as abstention reported in appendix).

Decision rule:

- $\kappa \geq 0.6$ overall *and* ≥ 0.5 per stratum $\rightarrow$ LLM judge accepted, no re-grading needed; paper reports κ.
- $\kappa < 0.5$ on any stratum $\rightarrow$ that stratum's grades switched to 3-judge majority vote (LLM-judge plus 2 different-prompted LLM re-runs) and the affected analyses are re-run with the new grades.
- Disagreement cells listed in appendix with one-line explanation, regardless of κ value.

4. Sample sizes and power

Study	Cells	n per condition	Power note
6	96 new + 48 existing = 144	24 per condition (false only)	At $\pi_1 = 0.875$, $\pi_2 = 0.$
7	144	36 per arm	Detects a 15-pt lift at
8	240	40 per condition (across 8 problems, 5 runs each)	Detects a 15-pt per-p
HJ	120	20–30 per stratum	$\kappa = 0.6$ detectable ag

5. Stopping rules

- **No early stopping** for any study. Each study is run to its pre-registered N before any per-condition statistics are computed beyond raw catch counts.
- **No N inflation** after the fact. If a study's effect appears marginal, the result is reported as marginal, not topped up.
- **One pre-registered re-grading round** allowed for the human judge audit only, triggered by $\kappa < 0.5$ on any stratum, and limited to 3-judge LLM majority vote on the affected cells.

6. Files this pre-registration governs

- Driver scripts:

- Study 6: `experiments/false_claim/run_false_claim.py` (re-invoked with `--n-additional 4`)
- Study 7: `experiments/inline_content/run_inline_content.py` (to be created) plus `experiments/inline_content/path_only_wrapper.py` — the thin wrapper that strips inline image content blocks from each `math-viz` tool result before the agent sees it. The baseline arm calls upstream `math-viz` directly; the manipulated arm routes through the wrapper.
- Study 8: `experiments/geometric_corpus/run_geometric.py` (to be created)

- Corpora:
  - Study 8: `experiments/geometric_corpus/excerpts/`, `experiments/geometric_corpus/cano`
- Judge audit:
  - Sampling: `experiments/judge_audit/sample.py`
  - Grading CLI: `experiments/judge_audit/grade.py`
  - Output: `experiments/judge_audit/sample.csv`, `experiments/judge_audit/grades.csv`
- Coding rubric for H7b: `experiments/inline_content/coding_rubric.md` (committed before any hand-coding).

7. Evidence trail

This pre-registration relies on the in-repo git history rather than a third-party timestamping service. The relevant invariants:

- `paper/pre_registration.md` was committed at `5eb78a4` (2026-05-21) before any Study 6 / 7 / 8 cells existed.
- Study-6-affecting revisions to this file (none expected) must precede the first Study 6 cell's `meta.json` timestamp; same rule for Studies 7 and 8.
- The author commits to not rewriting commits that touch this file after data collection begins for the corresponding study; verifying this is one `git log -- paper/pre_registration.md` away.

This is a weaker guarantee than an OSF or arXiv timestamp (the author controls the repo). A reviewer who needs stronger evidence can still fetch the file at any historical commit via the public GitHub mirror of this repository.